

Exercise A:

```
1. orders_exam:
CREATE TABLE orders_exam(
  id_order number(6) PRIMARY KEY,
  time_order date NOT NULL
);

2. Products_exam:
CREATE TABLE Products_exam(
  id_product number(6) PRIMARY KEY,
  name_product CHAR(60) NOT NULL,
  price NUMBER(8,2) NOT NULL
);

3. List_exam:
CREATE TABLE List_exam(
  id_list number(6) PRIMARY KEY,
  id_order number(6),
  id_product number(6),
  number_of_products NUMBER(6) NOT NULL,
  CONSTRAINT fk_id_order FOREIGN KEY (id_order) REFERENCES orders_exam(id_order),
  CONSTRAINT fk_id_product FOREIGN KEY (id_product) REFERENCES
products_exam(id_product)
);

4. orders_exam
INSERT ALL INTO orders_exam
VALUES(1, '14-JUN-2022')
INTO orders_exam VALUES(2, '15-JUN-2022')
INTO orders_exam VALUES(3, '18-JUL-2022')
SELECT 1 FROM DUAL;

INSERT ALL INTO Products_exam
VALUES(1, 'Intel Core i9-12900K', 559.99)
INTO Products_exam VALUES(2, 'Motherboard ASRock', 199.99)
INTO Products_exam VALUES(3, 'Seagate Expansion Portable', 59.99)
INTO Products_exam VALUES(4, 'Toshiba Canvio Basics', 57.99)
INTO Products_exam VALUES(5, 'Cooler Master Hyper', 59.99)
INTO Products_exam VALUES(6, 'Enermax AQUAFUSION 240', 89.99)
SELECT 1 FROM DUAL;

List_exam
INSERT ALL INTO List_exam
VALUES(1, 1, 1, 1)
INTO List_exam VALUES(2, 2, 2, 1)
INTO List_exam VALUES(3, 2, 3, 2)
INTO List_exam VALUES(4, 1, 3, 3)
INTO List_exam VALUES(5, 3, 5, 2)
INTO List_exam VALUES(6, 2, 6, 2)
SELECT 1 FROM DUAL;
```

Exercise B:

1. `SELECT MIN(price)
FROM Products_exam;`
2. `SELECT id_product, name_product, price, price+price*0.15
FROM Products_exam
WHERE price>400;`
3. `SELECT UPPER(name_product)
FROM Products_exam;`
4. `SELECT number_of_products , id_product, id_order
FROM List_exam;`
5. `SELECT name_product, price
FROM Products_exam;`
6. `SELECT name_product || ' costs ' || price || ' euro.'
FROM Products_exam;`
7. `SELECT min(name_product)
FROM Products_exam
ORDER BY name_product;`
8. `CREATE VIEW viewB008 AS
SELECT *
FROM Products_exam;`
9. `SELECT *
FROM List_exam JOIN orders_exam USING (id_order);`
10. `SELECT id_order, id_product, name_product, price
FROM List_exam JOIN products_exam USING (id_product);`
11. `SELECT *
FROM List_exam JOIN orders_exam USING (id_order)
WHERE time_order BETWEEN '01-JUL-2022' AND '31-JUL-2022'`
12. `SELECT id_order, id_product, name_product, price
FROM List_exam NATURAL JOIN products_exam;`
13. `SELECT *
FROM List_exam NATURAL JOIN products_exam NATURAL JOIN orders_exam
WHERE ID_LIST=6;`
14. `UPDATE Products_exam
SET PRICE = 429.75
WHERE name_product = 'Intel Core i9-12900K';`

15. UPDATE Products_exam
SET PRICE = (SELECT price FROM Products_exam WHERE name_product = 'Enermax
AQUAFUSION 240')
WHERE name_product = 'Toshiba Canvio Basics';
16. SELECT *
FROM List_exam NATURAL JOIN products_exam NATURAL JOIN orders_exam;
17. SELECT id_product, name_product, number_of_products
FROM List_exam JOIN products_exam Using(id_product)
WHERE id_order = 1;
18. CREATE VIEW viewB018 AS
SELECT *
FROM Products_exam
WHERE price < 100;
19. SELECT PRICE
FROM Products_exam NATURAL JOIN List_exam
WHERE ID_ORDER =2;
20. SELECT PRICE, name_product
FROM Products_exam
WHERE PRICE > (SELECT PRICE FROM Products_exam WHERE name_product =
'Seagate Expansion Portable');

Part 2

1. SELECT first_name, last_name, hire_date
FROM Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Matos');
2. SELECT first_name, last_name, hire_date
FROM Employees
WHERE hire_date > (SELECT hire_date FROM Employees WHERE last_name = 'Rajs');
3. SELECT first_name, last_name, salary, salary+500
FROM Employees
WHERE hire_date > (SELECT hire_date FROM Employees WHERE last_name = 'Matos');
4. SELECT first_name, last_name, hire_date
FROM Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Rajs' and
hire_date > (SELECT hire_date FROM Employees WHERE last_name = 'De Haan'));
5. SELECT first_name, last_name, location_id
FROM Employees NATURAL JOIN locations
WHERE salary > (SELECT salary FROM Employees WHERE last_name = 'Grant')
ORDER BY hire_date;
6. SELECT first_name, last_name, salary
FROM Employees
WHERE manager_id = (SELECT manager_id FROM Employees WHERE last_name =
'Vargas');
7. SELECT first_name, last_name, salary, start_date, end_date
FROM Employees JOIN job_history USING(department_id)
WHERE first_name = 'G%';
8. SELECT street_address, city, postal_code
FROM Employees NATURAL JOIN locations
WHERE department_id = (SELECT department_id FROM Employees WHERE last_name
= 'Rajs');
9. SELECT first_name, last_name, salary, DEPARTMENT_NAME, LOCATION_ID
FROM DEPARTMENTS NATURAL JOIN Employees
WHERE manager_id = (SELECT manager_id FROM Employees WHERE last_name =
'Matos');
10. SELECT first_name, last_name, salary, DEPARTMENT_NAME, LOCATION_ID
FROM DEPARTMENTS NATURAL JOIN Employees
WHERE manager_id = (SELECT manager_id FROM Employees WHERE last_name =
'Vargas')
ORDER BY salary DESC;

11. SELECT first_name, salary
FROM DEPARTMENTS NATURAL JOIN Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Vargas' or
last_name = 'Taylor');
12. SELECT ROUND(AVG(salary), 2)
FROM Employees
WHERE manager_id = (SELECT manager_id FROM Employees WHERE last_name =
'Matos');
13. SELECT UPPER(last_name) || ' works in ' || LOWER(department_name)
FROM DEPARTMENTS NATURAL JOIN Employees;
14. SELECT first_name, last_name, salary, department_name
FROM DEPARTMENTS NATURAL JOIN Employees
WHERE salary > (SELECT min(salary) FROM Employees WHERE hire_date >
(SELECT hire_date FROM Employees WHERE last_name = 'Rajs'));
15. SELECT first_name, last_name, salary
FROM Employees
WHERE salary > (SELECT avg(salary) FROM Employees WHERE department_id = 50);
16. SELECT first_name, last_name, salary, department_name, LOCATION_ID
FROM Employees NATURAL JOIN locations NATURAL JOIN DEPARTMENTS
WHERE salary < (SELECT avg(salary) FROM Employees WHERE hire_date <
(SELECT hire_date FROM Employees WHERE last_name = 'Matos'));
17. SELECT *
FROM Employees NATURAL JOIN locations NATURAL JOIN DEPARTMENTS
WHERE hire_date < (SELECT hire_date FROM Employees WHERE last_name = 'Vargas');
18. SELECT *
FROM locations NATURAL JOIN DEPARTMENTS NATURAL JOIN Employees NATURAL
JOIN job_history
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Higgins');
- 18.2. SELECT *
FROM locations JOIN DEPARTMENTS USING(location_id) JOIN Employees
USING(department_id) JOIN job_history USING(department_id)
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Higgins');
19. SELECT UPPER(first_name) || ' ' || UPPER(last_name), salary, department_ID,
MANAGER_ID, start_date, end_date
FROM Employees JOIN job_history USING(department_id);
20. SELECT e.*
FROM DEPARTMENTS d JOIN Employees e ON d.department_id = e.department_id
WHERE department_name = 'Shipping' or department_name = 'Executive';

21. SELECT first_name, last_name, salary
FROM Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Vargas')
ORDER BY HIRE_DATE;
22. SELECT first_name, last_name, hire_date
FROM Employees
WHERE department_id = (SELECT department_id FROM Employees WHERE last_name = 'Mourgos') AND hire_date > (SELECT hire_date FROM Employees WHERE last_name = 'Abel');
23. SELECT first_name, last_name, department_id
FROM Employees
WHERE salary > (SELECT salary FROM Employees WHERE last_name = 'Mourgos')
ORDER BY hire_date;
24. CREATE VIEW viewc024 AS
SELECT first_name, last_name, salary, hire_date
FROM Employees
WHERE hire_date > (SELECT hire_date FROM Employees WHERE last_name = 'Grant');
25. CREATE VIEW viewc025 AS
SELECT first_name, last_name, manager_id, salary
FROM Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Taylor');
26. SELECT first_name, last_name, salary, hire_date
FROM Employees
ORDER BY salary;
27. SELECT first_name, last_name, salary, hire_date
FROM Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Matos')
ORDER BY salary DESC;
28. SELECT first_name, last_name, salary, hire_date
FROM Employees
WHERE department_id = (SELECT department_id FROM Employees WHERE last_name = 'Matos') and salary > 8000
ORDER BY last_name DESC;
29. SELECT first_name, last_name, salary
FROM Employees
WHERE department_id = (SELECT department_id FROM Employees WHERE last_name = 'Lorentz');
30. SELECT first_name, last_name, salary
FROM Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Higgins');

31. SELECT first_name, last_name, salary
FROM Employees
WHERE salary < (SELECT MAX(salary) FROM Employees WHERE manager_id=149);
32. SELECT first_name || ' ' || last_name AS "Name"
FROM Employees
WHERE job_id = (SELECT job_id FROM Employees WHERE last_name = 'Lorentz');
33. SELECT e.*
FROM DEPARTMENTS d JOIN Employees e ON d.department_id = e.department_id
WHERE e.department_id != (SELECT department_id FROM Employees WHERE
last_name = 'Rajs')
ORDER BY e.last_name;
34. SELECT e.*
FROM DEPARTMENTS d JOIN Employees e ON d.department_id = e.department_id
WHERE e.department_id = (SELECT department_id FROM Employees WHERE
last_name = 'Matos')
ORDER BY e.hire_date;
35. SELECT first_name, last_name, salary
FROM Employees
WHERE job_id = 'AD_VP' or job_id = 'SA_REP'
ORDER BY hire_date;